

AFFORDABLE NEW JERSEY

Comfort Partners has helped more than 27,000 New Jersey households better manage their utility costs by saving energy.

by NEAL GALE AND JOHN AUGUSTINO

When you're short on money, sometimes it takes a little creativity to get by. In Neptune, New Jersey, a woman living in a small single-family home that was so ravaged by termites that the walls were bare heated water on her stove—for two years—after her water heater stopped working. Her gas consumption from heating rose to 1,450 therms (more than 100,000 BTU/ft²). This home was clearly beyond the scope of a typical weatherization program.

Another 85-year-old woman in an old three-story Victorian in Irvington, New Jersey, needed a way to stay warm when her gas-fired boiler quit. Her solution was to buy four electric space heaters and use a gas oven for heat. This unorthodox heating method was elevating CO in the house to dangerous levels, and testing of the boiler revealed a cracked heat exchanger. Meanwhile, her space heaters were causing electric use in the house to exceed 18,600 kWh annually (18,000 kWh is about 20% above average for New Jersey).

"People, regardless of their income status, should not have to sacrifice

comfort and safety in their homes," says Maria Frederick, working group convenor for New Jersey Comfort Partners. "These are basic needs that can greatly influence a person's day-to-day life. We have programs in this country that help low-income people pay for necessities like food. We need similar programs that can help them have a decent and comfortable place to live."

Gas, and South Jersey Gas. The collaboration and commitment of these organizations has led to a large-scale program that remains unmatched in the industry. To date, Comfort Partners has helped more than 27,000 income-eligible households better manage their utility costs by conserving energy.

The program, which addresses all energy sources at once, helps participants achieve energy affordability through a



A Honeywell technician monitors combustion efficiency with a flue gas analyzer as part of an in-home energy audit.

Comfort Partners

Like many of their regulated counterparts across the country, each of the investor-owned gas and electric utilities in New Jersey had an energy affordability initiative to help low-income customers. In 2001, under the direction of the New Jersey Board of Public Utilities (BPU), the utilities joined together to design a single, comprehensive energy conservation program for all low-income residents statewide. The result: New Jersey Comfort Partners.

The Comfort Partners program is comanaged by seven utilities: Public Service Enterprise Group (PSE&G), Jersey Central Power and Light, Atlantic City Electric, Rockland Electric, Elizabethtown Gas, New Jersey Natural

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combination of energy education and the installation of measures that simultaneously improve building thermal performance and baseload efficiencies. The gas and electric companies split the costs according to energy usage, allowing the contractor to use the joint funds in ways that best serve the customer.

In the case of the Irvington woman, in addition to replacing her inefficient refrigerator, repairing or replacing several leaky windows, and installing a water heater blanket, pipe insulation, and weatherization measures such as door sweeps and weather stripping, the Comfort Partners program replaced her boiler, not only saving energy but also potentially saving her life. The gas and electric utilities split the bill 36% to 63%, based on her high electric consumption.

At the house in Neptune, the program implementers quickly realized that solving the problem would require more resources than they could provide. So they coordinated efforts with the sponsoring utilities and four separate community-based organizations to repair termite damage, reinsulate and Sheetrock walls, replace windows and doors, and correct several safety issues. The utility-focused work split 89% for gas and 11% for electric, based on the owner's extremely high gas use.

Even more significantly, Comfort Partners helped leverage more than double the program expenditures from other, outside agencies to bring more than \$20,000 in improvements to a very critical situation. The woman now lives in a safe, efficient home.

"This is the only large-scale utility-managed program that addresses both fuel sources," Frederick says. "We realize savings by sharing costs and sending one contractor to serve customers of two utilities."

Statewide Coordination Posed Challenges

In the fall of 2005, the American Council for an Energy-Efficient Economy (ACEEE) recognized Comfort Partners as an exemplary low-income

energy efficiency program. To merit this award from one of the industry's most recognized professional organizations, the program had to overcome major challenges.

The program implementers encountered a wide range of housing types and conditions throughout the state. These included many deteriorated, aging structures, which presented some of the most pressing needs. "It was clear we needed outside partners with a high level of specialized knowledge and expertise," Frederick says. "It would have been too costly to train our own staff for something this comprehensive and far-reaching."

Comfort Partners chose to work with Honeywell, our company. Honeywell had helped manage similar programs for four of the participating utilities. And we were able to develop a methodology for applying the program design to meet the needs of a wide range of building types.

As the primary contractor for Comfort Partners, Honeywell Utility Solutions assisted with the design and launch of the program, providing individualized services to meet the unique support needs of each partner utility. This allowed all member utilities to provide the same level of service to their customers.

Honeywell now enlists customers; determines which customers have the greatest savings opportunity or need; conducts full-site evaluations of residences; provides customer energy education; and installs conservation measures through an experienced staff and team of subcontractors. Additional duties include training Honeywell and subcontractor staff, collecting data for evaluation, coordinating services with community-based organizations, and providing tips and additional resources to help customers better keep up with monthly payments.

"It's the same challenge that faces every other business: How do you efficiently maximize resources and services?" says Tom McMahon, residential program manager for Honeywell Utility Solutions. "In our case, that means maxi-

mizing the number of cost-effective energy-saving measures and services that can be delivered with program funds. The end goal is to benefit the greatest number of customers in the most expeditious manner possible."

A Model of Best Practices

"Historically, energy audits were prescriptive. Energy-saving measures were predetermined and not necessarily based on the specific needs of the house," McMahon says. "With Comfort Partners, we take a whole-house approach and address the unique needs of each household." The process begins with a comprehensive audit of the whole house to determine all energy use and savings opportunities. Using modern diagnostic techniques, highly trained technicians individually assess each home and determine which measures will have the most impact on energy efficiency.

The technicians then develop an energy reduction work plan. Some of the measures include upgrading insulation in the attic or walls, sealing cracks, fixing leaks in the duct systems, and replacing inefficient refrigerators. Other services and measures include providing efficient lighting products, taking steps to conserve hot water, installing new thermostats, and maintaining heating and cooling equipment. In addition, the technicians review, test, and correct a wide range of health and safety concerns. On average, program services are delivered within 30 days of the initial audit.

Because Comfort Partners addresses both gas and electric consumption, the measures implemented on one side often help to solve problems on the other. For example, air sealing a house to reduce electric A/C costs will help to reduce the cost of gas heating as well. "The utilities involved with each customer project get 100% of the benefit, but don't pay 100% of the costs," McMahon says. "Also, if you reduce the electric bill, the customer has more money to pay the gas bill and vice versa." This combined approach helped bring together the seven New Jersey utilities to offer

a uniform service and has been a key factor in the program's success.

Savings Support Larger Goals

In the case of the Irvington woman, not only did the measures installed increase her health and comfort, but they reduced her gas usage by nearly 20%. (Electric savings were negligible, due to use of extensive medical equipment.)

As for the Neptune customer, although the house showed no appreciable gas savings, electric usage declined 29% the first year, while the woman's comfort level increased significantly. In both of these cases, energy savings was not the primary benefit. The primary benefit was the increased health—or perhaps the saved lives—of the occupants. However, Comfort Partners is having a significant impact on energy use—and on the environment—statewide.

In 2005, 6,400 participants received much-needed energy-saving measures and services at a program cost of \$14 million. An impact analysis for program work performed in these homes revealed significant savings. These included

- a total estimated annual electric savings of 5.7 million kWh;
- a total estimated annual gas savings of 48,730 therms;
- a total projected lifetime electric savings of 106.4 million kWh; and
- a total projected lifetime gas savings of 18.5 million therms.

Reducing the energy burden on those low-income customers who have the most difficulty paying will also help keep them from falling behind on payments. Not only does that avoid a shut-off for the customer, but it also saves the utility time and money. The utility does not have to send collection letters, does not have to pay someone to shut off the energy, and does not have to absorb the cost of reconnecting the customer.

And through the program's health and safety diagnostics, hundreds of customers have been saved from CO poisoning and other life-threatening health risks. The program has identified

and corrected scores of cracked heat exchangers, blocked chimneys, undetected gas leaks, unvented appliances, unhealthy moisture levels, and many other potentially life-threatening situations.

In all, thousands of at-risk residents have received life-improving services through this vital program. And at the same time, the program has reduced the environmental impact of burning fossil fuels and the long-term social costs of the resulting environmental degradation for all of New Jersey.

Besides the energy savings, the program contributes to the state's drive to reduce pollutants through cleaner new sources and more efficient use of existing sources of energy. Extrapolating from the estimated energy savings, program services through the end of 2005 have cut CO₂ emissions by more than 117 million lb. In clean-air terms, this reduction is equivalent to taking 11,700 cars off the road or planting 15,900 acres of trees.

New Jersey's Comfort Partners program also integrates the participant outreach effort with the statewide Universal Service Fund (USF), which provides low-income residents with financial assistance for gas and electric bills. Not only do Honeywell field technicians and customer service representatives provide services within the scope of Comfort Partners, but they also link participants to a host of other assistance resources—from repair loans and emergency payments to childcare assistance and other social services. The program works cooperatively with a statewide network of community-based organizations to coordinate delivery of multiple assistance resources.

Energy Costs Drive Demand

Rising energy costs have led to a heightened interest in reducing energy use on the part of consumers struggling to pay their bills. They have also led program sponsors to seek ways to serve more people. "The need is still great, so we're hoping to serve 7,500 new customers in 2006," Frederick says. "The board [BPU] has been generous in pro-

viding funding for the program because they see how effective it is in helping customers save energy." In 2005, the utilities put out a request for proposals to expand program resources, bringing two more implementation contractors, CMC Services, Incorporated, (see "Upgrading Energy Technology—A Free-Market Solution," *HE* Jan/Feb '06, p. 8) and EIC Comfort Homes, into the program. Program surveys confirm consistently high customer satisfaction with the Comfort Partners services.

"We've already helped tens of thousands of people, yet we have much to look forward to," says Frederick. "The Comfort Partners program promises to deliver a positive impact on the lives of many more New Jersey residents in the months and years to come."



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For more information:

Contact New Jersey Comfort Partners at (1-888)773-8326, or access the New Jersey Clean Energy Web site, www.njcleanenergy.com, for information on Comfort Partners and all of New Jersey's energy efficiency programs.

Contact Honeywell Utility Solutions at (1-800)345-6770, Ext. 615, or www.honeywell.com/utility.